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EV CHARGING STATION NETWORK MANAGEMENT SYSTEM

GROUP 28

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1. Introduction

The rapid expansion of electric vehicle (EV) adoption has created a growing demand for reliable, accessible, and efficiently managed charging infrastructure. As EV ownership increases, both drivers and station operators face challenges related to locating available stations, managing charging sessions, handling payments, and maintaining hardware. A centralized software system that addresses these challenges is essential for the sustainable growth of the EV ecosystem.

The EV Charging Station Network Management System is a web-based platform designed to manage a network of electric vehicle charging stations, support reservation and payment operations, and monitor charging sessions in real time. The system is analyzed from eight viewpoints. EV Drivers use the system to search for stations, register vehicles, make reservations, and monitor charging sessions. Station Operators manage charger availability, station operations, and maintenance-related issues. Administrators oversee users, stations, operators, and reporting functions. The Map and Navigation Service supports station discovery, geolocation, distance calculation, and route guidance. The Payment / Wallet Service handles wallet top-ups, balance checking, fee deduction, refunds, and receipt generation. In addition, the Security Department introduces requirements for authentication, authorization, audit logging, and protection of sensitive data; Government Regulations impose requirements related to transaction transparency, record retention, and lawful data handling; and UI Standards define interface consistency, feedback, validation, and usability expectations across the platform.

1.1 Purpose

The purpose of this document is to present a comprehensive requirements analysis for the EV Charging Station Network Management System. This report identifies all stakeholder viewpoints, defines and classifies requirements and models the problem domain. It serves as the foundational artifact upon which subsequent design, implementation, and testing phases will be built.

1.2 Scope

The system encompasses the following core capabilities:

- **Station Catalog & Charger Types:** A database of charging stations including location details, charger types (AC/DC), power outputs (22kW, 50kW, 150kW), pricing per kWh, connector types (Type 2, CCS, CHAdeMO), and operating hours.
- **Interactive Map & Station Finder:** Google Maps API integration to display stations on an interactive map with real-time availability status (Available/Occupied/Offline), filtering, distance calculation, and navigation.
- **Vehicle Registration:** Registration of electric vehicles with brand, model, battery capacity, connector type, and plate number. The system validates vehicle–charger compatibility during reservation.
- **Charging Reservation & Conflict Handling:** Time-slot-based reservation with business rules (maximum 2-hour sessions, no bookings more than 24 hours in advance) and double-booking prevention.
- **Online Payment & Wallet System:** Secure wallet top-ups, automatic fee deduction based on kWh consumed, refund processing, and digital receipt generation.
- **Charging Session Tracking:** Real-time monitoring of active sessions including current kWh, estimated time remaining, and running cost.
- **Station Maintenance & Issue Reporting:** User-submitted issue reports, operator-managed service status, automatic cancellation of affected reservations, and user notification.

- **Favorites & Charging History:** Saved favorite stations and complete charging history with energy consumed and cost details.
- **Notifications:** Alerts for reservation confirmations, charging status, low wallet balance, and station availability changes.
- **Administrative Reporting:** Revenue reports, station utilization statistics, peak-hour analysis, and user activity summaries.

1.3 Definitions, Acronyms, and Abbreviations

TERM	DEFINITION
EV	Electric Vehicle
AC	Alternating Current
DC	Direct Current
KW	Kilowatt (unit of power)
KWH	Kilowatt-hour (unit of energy)
CCS	Combined Charging System (connector type)
CHADEMO	CHArge de MOve (DC fast-charge connector standard)
TYPE 2	IEC 62196 Type 2 connector (EU standard)
SOC	State of Charge (battery percentage)
API	Application Programming Interface
RBAC	Role-Based Access Control

1.4 Methodology

This analysis follows the **Waterfall Process Model** as required by the course. The viewpoint-oriented approach is used for requirements elicitation, ensuring all stakeholder perspectives are captured. Requirements are classified both by functionality (functional, non-functional, domain) and by lifetime (volatile, enduring). The MoSCoW prioritization method is employed for requirement negotiation.

2. Identification of Viewpoints

2.1 Principal Viewpoints of the System

	VIEWPOINT	TYPE	DESCRIPTION
VP-1	EV Driver	Direct User	The primary end-user who searches for stations, registers vehicles, makes reservations, charges vehicles, and manages payments.
VP-2	Station Operator	Direct User	Manages physical charging stations, handles maintenance, and monitors charger status.
VP-3	Administrator	Direct User	Oversees the entire charging network, manages users, configures system rules, and generates reports.
VP-4	Map and Navigation Service	External System	External mapping service (Google Maps API) providing interactive maps, geolocation, distance calculation, and navigation.
VP-5	Payment / Wallet Service	External System	Payment processing and digital wallet service that handles top-ups, automatic deductions, refunds, and financial security.

VP-6	UI Standards	Domain	Defines interface consistency, usability, and design principles across the platform to ensure a predictable user experience.
VP-7	Government Regulations	Domain	Represents legal and compliance constraints regarding financial transparency, record retention, and data handling.
VP-8	Security Department	Indirect	Captures organizational concerns related to authentication, authorization, audit logging, and protection of sensitive data.

2.2 Description of Each Viewpoint

VP-1: EV Driver

The EV Driver is the primary end-user of the mobile and web application. Their central goal is to charge their vehicle conveniently. They interact with the system to search for nearby stations on an interactive map, check real-time availability, verify connector compatibility with their registered vehicle, create and manage reservations, pay via a digital wallet, and monitor live charging progress. EV Drivers also contribute to platform quality by reporting station issues, saving favorite stations for quick access, and reviewing their complete charging history.

Concerns: Ease of use, station availability, compatibility assurance, cost transparency, reliable notifications, quick access to nearby stations.

VP-2: Station Operator

Station Operators are accountable for the physical infrastructure. Their primary concerns are charger uptime, revenue generation, and responsive maintenance. They interact with the system through an operator dashboard to monitor charger statuses, configure stations and chargers, set pricing per kWh, mark equipment out of service during maintenance which automatically cancels affected reservations and notifies users review issue reports submitted by drivers, and track utilization statistics for their stations.

Concerns: Operational efficiency, timely maintenance, accurate status reporting, revenue visibility, minimal downtime.

VP-3: Administrator

Administrators have the highest privilege level and manage the entire network. They add and remove charging stations, create and manage operator accounts, activate or deactivate user accounts, and generate aggregate reports including revenue breakdowns, station utilization statistics, peak-hour analysis, and user activity summaries. A system-wide dashboard provides them with key metrics such as total active sessions, station counts, and daily transaction volume.

Concerns: Network-wide visibility, data-driven decision making, system reliability, regulatory compliance.

VP-4: Map and Navigation Service

This viewpoint represents the external mapping service (Google Maps API) integrated into the platform. It is responsible for rendering an interactive map with all charging stations displayed as color-coded markers (green for Available, yellow for Occupied, red for Offline), determining the user's current position via the Geolocation API, calculating driving distances from the user to each station using the Distance Matrix API, and providing turn-by-turn navigation directions to the selected station via the Directions API. Station markers update in real time as availability changes.

Concerns: API availability, accurate geolocation, real-time data rendering, map performance.

VP-5: Payment / Wallet Service

The Payment / Wallet Service viewpoint represents the external financial gateway integrated for all monetary operations. It provides each user with a digital wallet where they can view their balance, top up via credit/debit card through a secure payment gateway, and review their transaction history. Charging fees are automatically deducted from the wallet based on kWh consumed upon session completion. The service processes refunds for eligible reservation cancellations, generates a digital receipt for every transaction, maintains a tamper-evident transaction log for audit purposes, and rejects charging session initiation when the wallet balance is insufficient. All transactions are processed over encrypted channels (HTTPS/TLS).

Concerns: Transaction security, processing reliability, compliance with financial regulations, low latency.

VP-6: UI Standards

The UI Standards viewpoint defines interface consistency, usability expectations, validation behavior, readable information presentation, and common design principles across the platform. It constrains how major user workflows such as vehicle registration, reservation, payment, and charging-session monitoring are presented so that the system remains understandable, predictable, and easy to use on the web interface.

Concerns: Interface consistency, usability, readability, clear feedback, effective validation, low learning curve.

VP-7: Government Regulations

This viewpoint represents legal and compliance-related constraints that affect the system's handling of transactions, receipts, financial records, and personal or payment-related data. It ensures that the platform supports transparent financial operations, lawful record retention, and auditable transaction management.

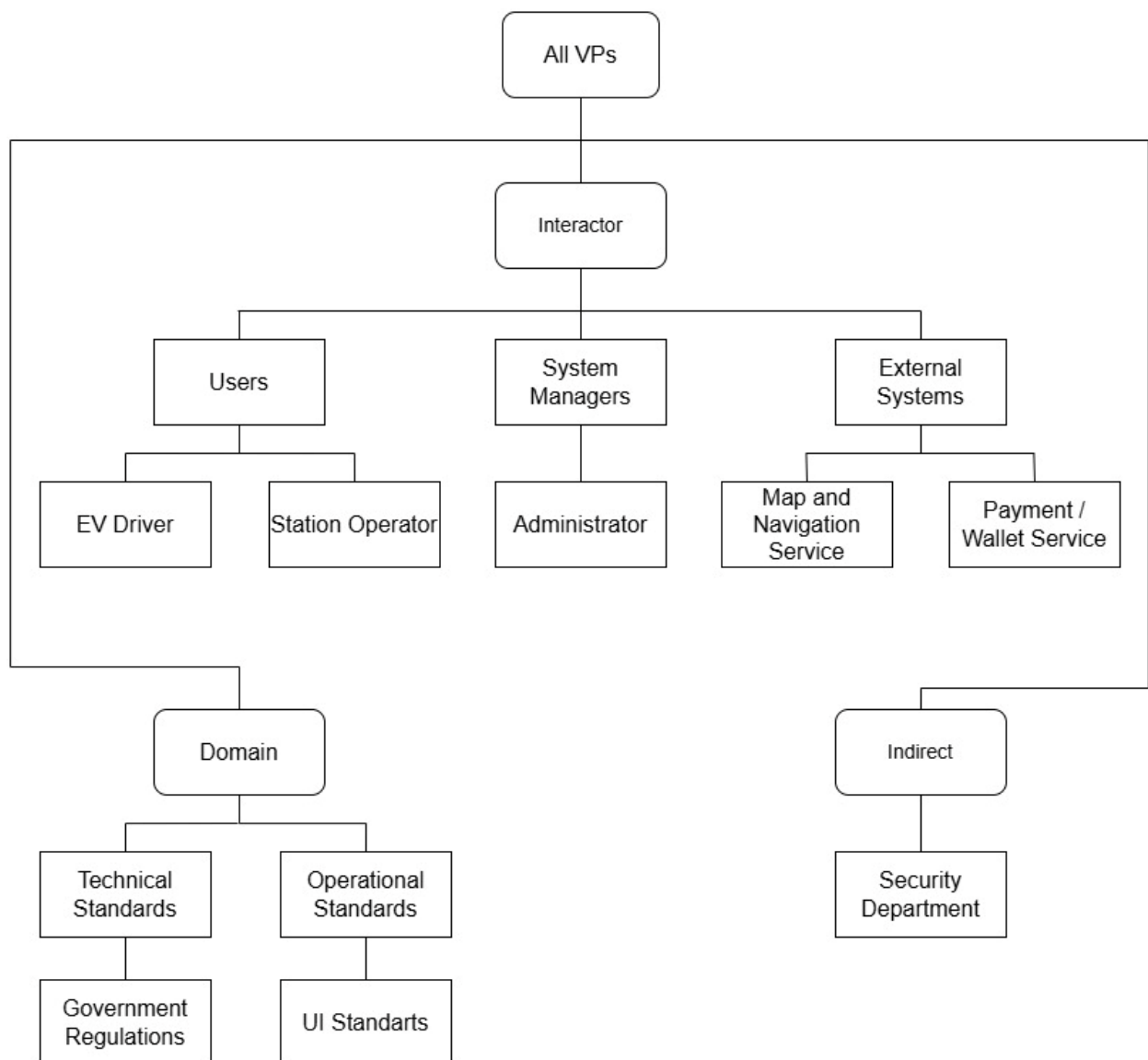
Concerns: Transaction transparency, legal compliance, record retention, financial traceability, lawful data handling.

VP-8: Security Department

The Security Department viewpoint captures organizational security concerns related to controlled access, auditability, and the protection of sensitive functions and data. It introduces requirements for authentication, authorization, logging of security-relevant actions, and detection of suspicious access behavior.

Concerns: Authentication, authorization, audit logging, protection of sensitive data, detection of suspicious access attempts.

2.3 Viewpoint Hierarchy Diagram



3. Requirements Definition

3.1 EV Driver

Requirements for the primary end-user who discovers stations, registers vehicles, makes reservations, monitors charging sessions, and manages payments.

ID	VIEWPOINT	DEFINITION OF REQUIREMENT
EDR-01	EV Driver	The system shall allow an EV driver to register one or more electric vehicles by entering brand, model, battery capacity (kWh), connector type (Type 2, CCS, or CHAdEMO), and plate number.
EDR-02	EV Driver	The system shall validate the completeness and correctness of vehicle information (supported connector type, unique plate number, battery capacity within 1–200 kWh) before saving the vehicle record.
EDR-03	EV Driver	The system shall allow the driver to view all charging stations on an interactive map with the driver's current location displayed.

EDR-04	EV Driver	The system shall allow the driver to filter charging stations by connector type, power output, and price, and view detailed station information including charger types (AC/DC), power output (22kW, 50kW, 150kW), pricing per kWh, and operating hours.
EDR-05	EV Driver	The system shall display the real-time availability status of charging stations and chargers using color-coded markers: Green (Available), Yellow (Occupied), Red (Offline).
EDR-06	EV Driver	The system shall allow the driver to select a station from the map and create a reservation for a specific charger, date, and time slot.
EDR-07	EV Driver	The system shall validate vehicle–charger compatibility by matching the registered vehicle's connector type against the selected charger's connector before confirming the reservation.
EDR-08	EV Driver	The system shall enforce a maximum session duration of 2 hours per reservation, allow reservations only within the selected station's operating hours, prevent reservations more than 24 hours in advance, and reject double-bookings for overlapping time slots.
EDR-09	EV Driver	The system shall allow the driver to monitor real-time charging session progress, including current kWh delivered, estimated time remaining, and running cost.
EDR-10	EV Driver	The system shall allow the driver to view their complete charging history, including station name, date/time, energy consumed (kWh), duration, and total cost.
EDR-11	EV Driver	The system shall allow the driver to save charging stations as favorites for quick future access.
EDR-12	EV Driver	The system shall allow the driver to report station or charger issues by selecting a category (e.g., "Connector damaged", "Payment terminal broken") and providing an optional description.
EDR-13	EV Driver	The system shall notify the driver about reservation confirmation, charging started, charging completed, low wallet balance, reservation cancellation due to maintenance, and station availability changes affecting favorited or reserved stations.
EDR-14	EV Driver	The system shall allow the driver to cancel an existing reservation before the scheduled start time and initiate a refund to the wallet.
EDR-15	EV Driver	The system shall calculate and display an estimated charging cost before reservation confirmation based on the selected charger's pricing and the requested reservation time slot.
EDR-16	EV Driver	If charging does not start within 15 minutes of the reservation start time, the system shall mark the reservation as No-Show, release the reserved slot, and notify the user.
EDR-17	EV Driver	The system shall allow the driver to set a target SoC percentage for a charging session, and the system shall automatically stop charging when the target SoC is reached.
EDR-18	EV Driver	The system shall preserve active charging session data during temporary network connectivity loss and synchronize cached session data within 30 seconds after connectivity is restored, without interrupting charging.

3.2 Station Operator

Requirements for operators who manage physical charging infrastructure, respond to reported issues, and monitor station performance.

ID	VIEWPOINT	DEFINITION OF REQUIREMENT
SOR-01	Station Operator	The system shall allow station operators to update the operational status of each charger as Available, Occupied, or Out of Service.
SOR-02	Station Operator	The system shall allow station operators to manage charging station information, including charger type, power output, pricing per kWh, and operating hours.
SOR-03	Station Operator	The system shall allow station operators to view current and upcoming reservations for their stations.
SOR-04	Station Operator	The system shall allow station operators to review issue reports submitted by users and update their resolution status (Open, In Progress, Resolved).
SOR-05	Station Operator	The system shall automatically cancel reservations affected by chargers marked as out of service and notify the affected users.
SOR-06	Station Operator	The system shall allow station operators to view charger utilization and station occupancy statistics for operational decision-making.

3.3 Administrator

Requirements for the platform administrator who governs all users, operators, stations, and network-wide policies.

ID	VIEWPOINT	DEFINITION OF REQUIREMENT
ADR-01	Administrator	The system shall allow administrators to add and remove charging stations from the network.
ADR-02	Administrator	The system shall allow administrators to create, update, and deactivate operator and user accounts.
ADR-03	Administrator	The system shall allow administrators to generate revenue reports and station utilization statistics.
ADR-04	Administrator	The system shall allow administrators to analyze peak charging hours across the network.
ADR-05	Administrator	The system shall allow administrators to view user activity summaries, reservation volumes, cancellations, and maintenance trends.
ADR-06	Administrator	The system shall enforce role-based access control so that EV drivers, station operators, and administrators can access only the functions permitted to their roles.
ADR-07	Administrator	The system shall maintain a tamper-evident audit log for all financial transactions and retain transaction records for at least 5 years.
ADR-08	Administrator	The system shall provide 99.5% monthly availability, excluding scheduled maintenance periods.

3.4 Map and Navigation Service

Requirements governing the integration with the external mapping service (Google Maps API) for station visualization, geolocation, distance calculation, and route navigation.

ID	VIEWPOINT	DEFINITION OF REQUIREMENT
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MNS-01	Map & Navigation	The system shall obtain and display the user's current location through Google Maps geolocation integration.
MNS-02	Map & Navigation	The system shall display all charging stations as markers on the interactive map.
MNS-03	Map & Navigation	The system shall use color codes on station markers to indicate charger availability status: Green (Available), Yellow (Occupied), Red (Offline).
MNS-04	Map & Navigation	The system shall calculate and display the driving distance between the user's current location and each charging station.
MNS-05	Map & Navigation	The system shall provide navigation directions from the user's current location to the selected charging station.
MNS-06	Map & Navigation	The system shall refresh station marker data on the map within 5 seconds when charger availability status changes.
MNS-07	Map & Navigation	The system shall update station search and filter results on the map within 2 seconds under normal load.

3.5 Payment / Wallet Service

Requirements governing the integration with the external financial gateway for wallet management, billing operations, and receipt generation.

ID	VIEWPOINT	DEFINITION OF REQUIREMENT
PWS-01	Payment / Wallet	The system shall allow users to top up their wallet through a secure payment gateway.
PWS-02	Payment / Wallet	The system shall automatically deduct charging fees from the user's wallet based on the amount of energy consumed and the station's pricing per kWh after the session is completed.
PWS-03	Payment / Wallet	The system shall generate a digital receipt after payment is completed.
PWS-04	Payment / Wallet	The system shall process refunds for cancelled reservations when applicable.
PWS-05	Payment / Wallet	The system shall inform the user when the wallet balance is insufficient for reservation confirmation or active charging-related payments.
PWS-06	Payment / Wallet	The system shall allow users to view their current wallet balance and complete transaction history (top-ups, charges, refunds).
PWS-07	Payment / Wallet	The system shall verify that the user's wallet balance is sufficient for the estimated reservation cost before confirming a reservation.
PWS-08	Payment / Wallet	If the actual charging cost reaches the user's available wallet balance during an active session, the system shall stop the charging session safely, deduct the available balance, generate a partial-session receipt, and notify the user.
PWS-09	Payment / Wallet	The system shall encrypt all payment-related communications using TLS 1.2 or higher.

3.6 UI Standards

Requirements governing interface consistency, usability, visual feedback, validation behavior, and common design principles across the system.

ID	VIEWPOINT	DEFINITION OF REQUIREMENT
UIR-01	UI Standards	The system shall use a consistent navigation structure, page layout logic, and interaction pattern across the main workflows of vehicle registration, station search, reservation, payment, and charging-session monitoring.
UIR-02	UI Standards	The system shall display a confirmation message after successful actions and an error message after failed actions during reservation, payment, and charging-session workflows.
UIR-03	UI Standards	The system shall validate required form fields before submission and shall indicate missing or invalid inputs to the user on the same page.
UIR-04	UI Standards	The system should follow Material Design principles for visual hierarchy, component consistency, and interaction design.
UIR-05	UI Standards	The system shall present reservation, payment, and charging-session information in a readable and consistent format across all relevant pages.

3.7 Government Regulations

Requirements governing legal compliance, transaction transparency, financial record retention, and the handling of personal and payment-related data.

ID	VIEWPOINT	DEFINITION OF REQUIREMENT
GRR-01	Government Regulations	The system shall generate a digital receipt after each completed wallet top-up, refund, or charging payment transaction.
GRR-02	Government Regulations	The system shall display the charged amount, deducted wallet balance, and refund result to the user after each financial operation.
GRR-03	Government Regulations	The system shall retain transaction records and digital receipts for audit and legal review purposes.
GRR-04	Government Regulations	The system shall maintain a retrievable history of wallet and charging transactions so that disputes, complaints, and refund reviews can be handled when necessary.
GRR-05	Government Regulations	The system shall handle personal and payment-related data in accordance with applicable legal and regulatory requirements.

3.8 Security Department

Requirements governing authentication, authorization, auditability, and protection of sensitive functions and data.

ID	VIEWPOINT	DEFINITION OF REQUIREMENT
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SDR-01	Security Department	The system shall require users to log in before accessing wallet functions, reservation management, charging-session monitoring, or administrative features.
SDR-02	Security Department	The system shall allow access to operator and administrator functions only if the authenticated user has the required role.
SDR-03	Security Department	The system shall record login attempts, account updates, station status changes, and payment-related actions in an audit log.
SDR-04	Security Department	The system shall notify the administrator when repeated failed login attempts are detected for the same account within a defined period.
SDR-05	Security Department	The system shall protect sensitive functions and data against unauthorized access.

4. Requirements Classification (Considering Functionality)

All requirements are classified as Functional (FR), Non-Functional (NFR), or Domain (DR).

ID	VIEWPOINT	DEFINITION OF REQUIREMENT	FUNCTIONALITY
EDR-01	EV Driver	Register one or more EVs with brand, model, battery, connector, plate	Functional
EDR-02	EV Driver	Validate vehicle info completeness and correctness before saving	Functional
EDR-03	EV Driver	View all stations on interactive map with current location	Functional
EDR-04	EV Driver	Filter stations by connector, power, price; view detailed station info	Functional
EDR-05	EV Driver	Display real-time availability with color-coded markers	Functional
EDR-06	EV Driver	Select station from map and create reservation for charger/date/time	Functional
EDR-07	EV Driver	Validate vehicle–charger compatibility before reservation	Domain
EDR-08	EV Driver	Enforce max 2h session, max 24h advance, prevent double-booking	Domain
EDR-09	EV Driver	Monitor real-time session progress (kWh, time, cost)	Functional
EDR-10	EV Driver	View complete charging history	Functional
EDR-11	EV Driver	Save stations as favorites	Functional
EDR-12	EV Driver	Report station/charger issues with category and description	Functional
EDR-13	EV Driver	Notifications: reservation, charging status, low balance, cancellation	Functional
EDR-14	EV Driver	Cancel reservation before start time and initiate refund	Functional
EDR-15	EV Driver	Calculate and display estimated charging cost before reservation confirmation	Functional
EDR-16	EV Driver	Mark no-show after 15 minutes, release slot, notify user	Functional

EDR-17	EV Driver	Set target SoC and auto-stop charging at target	Functional
EDR-18	EV Driver	Preserve and synchronize session data during temporary connectivity loss	Non-Functional
SOR-01	Station Operator	Update charger operational status	Functional
SOR-02	Station Operator	Manage station info: charger type, power, pricing, hours	Functional
SOR-03	Station Operator	View current and upcoming reservations	Functional
SOR-04	Station Operator	Review issue reports and update resolution status	Functional
SOR-05	Station Operator	Auto-cancel reservations on out-of-service; notify users	Functional
SOR-06	Station Operator	View charger utilization and occupancy statistics	Functional
ADR-01	Administrator	Add/remove charging stations from network	Functional
ADR-02	Administrator	Create, update, deactivate operator and user accounts	Functional
ADR-03	Administrator	Generate revenue reports and utilization statistics	Functional
ADR-04	Administrator	Analyze peak charging hours	Functional
ADR-05	Administrator	View user activity, reservation volumes, maintenance trends	Functional
ADR-06	Administrator	Enforce role-based access control by user role	Non-Functional
ADR-07	Administrator	Maintain tamper-evident audit log and retain financial records for 5 years	Non-Functional
ADR-08	Administrator	Provide 99.5% monthly availability excluding scheduled maintenance	Non-Functional
MNS-01	Map & Navigation	Obtain and display user's current location	Functional
MNS-02	Map & Navigation	Display all stations as markers on map	Functional
MNS-03	Map & Navigation	Color-coded markers for availability status	Functional
MNS-04	Map & Navigation	Calculate and display distance from user to stations	Functional
MNS-05	Map & Navigation	Provide navigation directions to selected station	Functional
MNS-06	Map & Navigation	Refresh map data when station availability changes	Non-Functional
MNS-07	Map & Navigation	Update station search and filter results on the map within 2 seconds under normal load	Non-Functional
PWS-01	Payment / Wallet	Wallet top-up through secure payment gateway	Functional
PWS-02	Payment / Wallet	Auto-deduct fees based on kWh consumed × station pricing	Functional
PWS-03	Payment / Wallet	Generate digital receipt after payment	Functional
PWS-04	Payment / Wallet	Process refunds for cancelled reservations	Functional
PWS-05	Payment / Wallet	Inform user when wallet balance is insufficient	Functional
PWS-06	Payment / Wallet	View wallet balance and transaction history	Functional
PWS-07	Payment / Wallet	Verify wallet balance is sufficient for estimated reservation cost before confirmation	Functional

PWS-08	Payment / Wallet	Safely stop session when wallet balance is exhausted, deduct available balance, issue partial receipt, notify user	Functional
PWS-09	Payment / Wallet	Encrypt all payment-related communications using TLS 1.2 or higher	Non-Functional
UIR-01	UI Standards	Use consistent navigation, page layout logic, and interaction patterns across main workflows	Non-Functional
UIR-02	UI Standards	Display confirmation and error messages during reservation, payment, and charging workflows	Functional
UIR-03	UI Standards	Validate required form fields and indicate missing or invalid inputs on the same page	Functional
UIR-04	UI Standards	Follow Material Design principles for visual hierarchy and interaction design	Non-Functional
UIR-05	UI Standards	Present reservation, payment, and charging-session information in a readable and consistent format	Non-Functional
GRR-01	Government Regulations	Generate a digital receipt after each completed financial transaction	Functional
GRR-02	Government Regulations	Display charged amount, deducted balance, and refund result after each financial operation	Functional
GRR-03	Government Regulations	Retain transaction records and digital receipts for audit and legal review	Domain
GRR-04	Government Regulations	Maintain retrievable wallet and charging transaction history for disputes and reviews	Functional
GRR-05	Government Regulations	Handle personal and payment-related data in accordance with legal and regulatory requirements	Domain
SDR-01	Security Department	Require login before access to protected wallet, reservation, charging, or administrative functions	Functional
SDR-02	Security Department	Allow operator and administrator access only to authenticated users with the required role	Functional
SDR-03	Security Department	Record security-relevant actions in an audit log	Functional
SDR-04	Security Department	Notify the administrator when repeated failed login attempts are detected	Functional
SDR-05	Security Department	Protect sensitive functions and data against unauthorized access	Non-Functional

5. Requirements Classification (Considering Lifetime)

All requirements are classified as Enduring (stable) or Volatile (likely to change).

ID	VIEWPOINT	DEFINITION OF REQUIREMENT	LIFETIME
EDR-01	EV Driver	Register one or more EVs with brand, model, battery, connector, plate	Enduring
EDR-02	EV Driver	Validate vehicle info completeness and correctness before saving	Enduring

EDR-03	EV Driver	View all stations on interactive map with current location	Enduring
EDR-04	EV Driver	Filter stations by connector, power, price; view detailed station info	Volatile
EDR-05	EV Driver	Display real-time availability with color-coded markers	Enduring
EDR-06	EV Driver	Select station from map and create reservation for charger/date/time	Enduring
EDR-07	EV Driver	Validate vehicle–charger compatibility before reservation	Enduring
EDR-08	EV Driver	Enforce max 2h session, max 24h advance, prevent double-booking	Volatile
EDR-09	EV Driver	Monitor real-time session progress (kWh, time, cost)	Enduring
EDR-10	EV Driver	View complete charging history	Enduring
EDR-11	EV Driver	Save stations as favorites	Enduring
EDR-12	EV Driver	Report station/charger issues with category and description	Enduring
EDR-13	EV Driver	Notifications: reservation, charging status, low balance, cancellation	Enduring
EDR-14	EV Driver	Cancel reservation before start time and initiate refund	Enduring
EDR-15	EV Driver	Calculate and display estimated charging cost before reservation confirmation	Enduring
EDR-16	EV Driver	Mark no-show after 15 minutes, release slot, notify user	Volatile
EDR-17	EV Driver	Set target SoC and auto-stop charging at target	Volatile
EDR-18	EV Driver	Preserve and synchronize session data during temporary connectivity loss	Enduring
SOR-01	Station Operator	Update charger operational status	Enduring
SOR-02	Station Operator	Manage station info: charger type, power, pricing, hours	Volatile
SOR-03	Station Operator	View current and upcoming reservations	Enduring
SOR-04	Station Operator	Review issue reports and update resolution status	Enduring
SOR-05	Station Operator	Auto-cancel reservations on out-of-service; notify users	Enduring
SOR-06	Station Operator	View charger utilization and occupancy statistics	Enduring
ADR-01	Administrator	Add/remove charging stations from network	Enduring
ADR-02	Administrator	Create, update, deactivate operator and user accounts	Enduring
ADR-03	Administrator	Generate revenue reports and utilization statistics	Enduring
ADR-04	Administrator	Analyze peak charging hours	Enduring
ADR-05	Administrator	View user activity, reservation volumes, maintenance trends	Enduring
ADR-06	Administrator	Enforce role-based access control by user role	Enduring
ADR-07	Administrator	Maintain tamper-evident audit log and retain financial records for 5 years	Enduring

ADR-08	Administrator	Provide 99.5% monthly availability excluding scheduled maintenance	Enduring
MNS-01	Map & Navigation	Obtain and display user's current location	Enduring
MNS-02	Map & Navigation	Display all stations as markers on map	Enduring
MNS-03	Map & Navigation	Color-coded markers for availability status	Enduring
MNS-04	Map & Navigation	Calculate and display distance from user to stations	Enduring
MNS-05	Map & Navigation	Provide navigation directions to selected station	Enduring
MNS-06	Map & Navigation	Refresh map data when station availability changes	Enduring
MNS-07	Map & Navigation	Update station search and filter results on the map within 2 seconds under normal load	Enduring
PWS-01	Payment / Wallet	Wallet top-up through secure payment gateway	Enduring
PWS-02	Payment / Wallet	Auto-deduct fees based on kWh consumed × station pricing	Volatile
PWS-03	Payment / Wallet	Generate digital receipt after payment	Enduring
PWS-04	Payment / Wallet	Process refunds for cancelled reservations	Enduring
PWS-05	Payment / Wallet	Inform user when wallet balance is insufficient	Enduring
PWS-06	Payment / Wallet	View wallet balance and transaction history	Enduring
PWS-07	Payment / Wallet	Verify wallet balance is sufficient for estimated reservation cost before confirmation	Volatile
PWS-08	Payment / Wallet	Safely stop session when wallet balance is exhausted, deduct available balance, issue partial receipt, notify user	Volatile
PWS-09	Payment / Wallet	Encrypt all payment-related communications using TLS 1.2 or higher	Enduring
UIR-01	UI Standards	Use consistent navigation, page layout logic, and interaction patterns across main workflows	Enduring
UIR-02	UI Standards	Display confirmation and error messages during reservation, payment, and charging workflows	Enduring
UIR-03	UI Standards	Validate required form fields and indicate missing or invalid inputs on the same page	Enduring
UIR-04	UI Standards	Follow Material Design principles for visual hierarchy and interaction design	Volatile
UIR-05	UI Standards	Present reservation, payment, and charging-session information in a readable and consistent format	Enduring

GRR-01	Government Regulations	Generate a digital receipt after each completed financial transaction	Enduring
GRR-02	Government Regulations	Display charged amount, deducted balance, and refund result after each financial operation	Enduring
GRR-03	Government Regulations	Retain transaction records and digital receipts for audit and legal review	Enduring
GRR-04	Government Regulations	Maintain retrievable wallet and charging transaction history for disputes and reviews	Enduring
GRR-05	Government Regulations	Handle personal and payment-related data in accordance with legal and regulatory requirements	Enduring
SDR-01	Security Department	Require login before access to protected wallet, reservation, charging, or administrative functions	Enduring
SDR-02	Security Department	Allow operator and administrator access only to authenticated users with the required role	Enduring
SDR-03	Security Department	Record security-relevant actions in an audit log	Enduring
SDR-04	Security Department	Notify the administrator when repeated failed login attempts are detected	Volatile
SDR-05	Security Department	Protect sensitive functions and data against unauthorized access	Enduring

Rationale for Volatile Classifications

EDR-04: Filter criteria may expand as new charger types, power levels, or pricing models are introduced to the market.

EDR-08: Session duration limits and advance booking windows are configurable business rules that may change based on operator needs and market conditions.

SOR-02: Pricing structures and station configurations change frequently in response to electricity costs and competitive pressures.

PWS-02: Billing logic may evolve as new pricing models (time-based, tiered) or payment methods are introduced.

EDR-16: No-show grace period is a configurable business rule that may change.

PWS-07: Reservation payment sufficiency policy may evolve into deposit or pre-authorization models.

PWS-08: Mid-session balance handling may change depending on future billing policies.

UIR-04: The selected design guideline may change over time due to branding decisions, evolving usability preferences, or framework changes.

SDR-04: Suspicious-access notification thresholds and alert policies may change as security practices mature.

7. Requirements Prioritization and Negotiation

7.1 Prioritization Method

The **MoSCoW** method is used:

- **Must Have (M):** Essential — system cannot function without these.
- **Should Have (S):** Important — adds significant value but not critical for launch.
- **Could Have (C):** Desirable — enhances experience but can be deferred.

7.2 Prioritization Table

ID	VIEWPOINT	REQUIREMENT (SUMMARY)	PRIORITY
EDR-01	EV Driver	Register one or more EVs	Must Have
EDR-02	EV Driver	Validate vehicle info before saving	Must Have
EDR-03	EV Driver	View stations on interactive map with location	Must Have
EDR-04	EV Driver	Filter stations; view detailed info	Must Have
EDR-05	EV Driver	Real-time availability with color-coded markers	Must Have
EDR-06	EV Driver	Create reservation for charger/date/time	Must Have
EDR-07	EV Driver	Vehicle–charger compatibility check	Must Have
EDR-08	EV Driver	Enforce max 2h, max 24h advance, no double-booking	Must Have
EDR-09	EV Driver	Real-time session monitoring	Must Have
EDR-10	EV Driver	View charging history	Should Have
EDR-11	EV Driver	Save favorite stations	Could Have
EDR-12	EV Driver	Report station/charger issues	Should Have
EDR-13	EV Driver	Notifications	Should Have
EDR-14	EV Driver	Cancel reservation and initiate refund	Must Have
EDR-15	EV Driver	Display estimated charging cost before reservation confirmation	Must Have
EDR-16	EV Driver	Mark reservation as no-show after 15 minutes	Should Have
EDR-17	EV Driver	Allow target SoC and auto-stop at target	Could Have
EDR-18	EV Driver	Preserve and synchronize session data during connectivity loss	Should Have
SOR-01	Station Operator	Update charger operational status	Must Have
SOR-02	Station Operator	Manage station info and pricing	Must Have
SOR-03	Station Operator	View reservations for stations	Should Have
SOR-04	Station Operator	Review and manage issue reports	Should Have
SOR-05	Station Operator	Auto-cancel + notify on out-of-service	Must Have
SOR-06	Station Operator	View utilization and occupancy stats	Could Have
ADR-01	Administrator	Add/remove stations from network	Must Have
ADR-02	Administrator	Manage operator and user accounts	Must Have
ADR-03	Administrator	Revenue reports and utilization stats	Must Have
ADR-04	Administrator	Peak-hour analysis	Could Have
ADR-05	Administrator	User activity and maintenance trends	Could Have
ADR-06	Administrator	Enforce role-based access control	Must Have
ADR-07	Administrator	Maintain tamper-evident audit logs and retention	Must Have
ADR-08	Administrator	Provide 99.5% monthly availability	Should Have
MNS-01	Map & Navigation	Display user's current location	Must Have

MNS-02	Map & Navigation	Display stations as map markers	Must Have
MNS-03	Map & Navigation	Color-coded availability markers	Must Have
MNS-04	Map & Navigation	Distance calculation user → station	Must Have
MNS-05	Map & Navigation	Navigation directions to station	Must Have
MNS-06	Map & Navigation	Refresh map on availability change	Should Have
MNS-07	Map & Navigation	Update map search/filter results within 2 seconds	Should Have
PWS-01	Payment / Wallet	Wallet top-up via payment gateway	Must Have
PWS-02	Payment / Wallet	Auto-deduct fees based on kWh × price	Must Have
PWS-03	Payment / Wallet	Generate digital receipt	Should Have
PWS-04	Payment / Wallet	Process refunds for cancellations	Should Have
PWS-05	Payment / Wallet	Warn on insufficient balance	Must Have
PWS-06	Payment / Wallet	View wallet balance and transaction history	Must Have
PWS-07	Payment / Wallet	Verify estimated-cost balance before reservation	Must Have
PWS-08	Payment / Wallet	Safely stop charging when wallet is exhausted	Should Have
PWS-09	Payment / Wallet	Encrypt payment communications with TLS 1.2+	Must Have
UIR-01	UI Standards	Consistent navigation, layout logic, and interaction patterns across main workflows	Should Have
UIR-02	UI Standards	Display confirmation and error messages during key workflows	Must Have
UIR-03	UI Standards	Validate form fields and show missing or invalid inputs on the same page	Must Have
UIR-04	UI Standards	Follow Material Design principles	Could Have
UIR-05	UI Standards	Present key reservation, payment, and charging information in a readable format	Should Have
GRR-01	Government Regulations	Generate a digital receipt after each completed financial transaction	Must Have
GRR-02	Government Regulations	Display charged amount, deducted balance, and refund result after financial operations	Must Have
GRR-03	Government Regulations	Retain transaction records and digital receipts for audit and legal review	Must Have
GRR-04	Government Regulations	Maintain retrievable financial history for disputes and reviews	Should Have
GRR-05	Government Regulations	Handle personal and payment-related data according to regulations	Must Have
SDR-01	Security Department	Require login before access to protected functions	Must Have
SDR-02	Security Department	Allow operator and administrator access only for authorized roles	Must Have
SDR-03	Security Department	Record security-relevant actions in an audit log	Should Have
SDR-04	Security Department	Notify administrator about repeated failed login attempts	Could Have
SDR-05	Security Department	Protect sensitive functions and data against unauthorized access	Must Have

7.3 Negotiation Notes

1. **Session duration limit (EDR-08):** Stakeholders discussed whether 2 hours is sufficient for slow AC charging. Decision: Keep 2-hour maximum as the default but make it configurable so administrators can adjust per station type.
2. **Advance booking window (EDR-08):** Operators requested extending to 48 hours. Decision: Maintain 24-hour default per project specification, with future configurability.
3. **Payment methods (PWS-01):** Users requested mobile wallet (e.g., Papara, Tosla) support. Decision: Credit/debit cards are the mandatory first implementation; mobile wallets can be added later.
4. **Favorites (EDR-11):** Classified as "Could Have", not essential for core workflow but requested by interviewed EV drivers for convenience.
5. **Issue reporting (EDR-12):** Classified as "Should Have", valuable for platform quality but the core charging workflow functions without it.
6. **Estimated cost before reservation (EDR-15, PWS-07):** Stakeholders preferred seeing a non-binding estimate before confirmation. Decision: Show an estimated cost and require sufficient wallet balance before reservation confirmation.
7. **Mid-session balance depletion policy (PWS-08):** Two alternatives were considered: allowing negative balance or safe session stop. Decision: Stop charging safely when available wallet balance is exhausted and generate a partial-session receipt.
8. **UI consistency baseline (UIR-01, UIR-04):** Stakeholders agreed that the first release should maintain a consistent interface across core workflows. Decision: consistent navigation and feedback are required, while strict Material Design adoption remains optional.
9. **Financial transparency and retention (GRR-01, GRR-03):** Stakeholders agreed that receipt generation and record retention are mandatory due to auditability and user trust concerns. Decision: both are classified as Must Have.
10. **Security logging and alerting (SDR-03, SDR-04):** Stakeholders agreed that audit logging is important for accountability, but repeated failed-login alerts can be introduced with simpler thresholds in later iterations. Decision: audit logging is Should Have and suspicious-access alerts are Could Have for the initial release.

9. Use Cases of the Main Scenarios

UC-1: Vehicle Registration and Compatibility Validation

Actors: EV Driver

Inputs: Vehicle brand (string), vehicle model (string), battery capacity in kWh (number), connector type (enum: Type 2, CCS, CHAdeMO), plate number (string, format: XX YYY XXXX)

Preconditions:

- The EV Driver is authenticated in the system.
- The EV Driver has access to the vehicle registration function.

Action Flow:

1. The authenticated EV Driver navigates to the Vehicle Registration screen.
2. The system displays the vehicle registration form.
3. The driver enters the vehicle details: brand, model, battery capacity, connector type, and plate number.
4. The system validates that all required fields are filled in.

5. The system validates that the connector type is one of the supported types (Type 2, CCS, CHAdeMO).
6. The system validates that the plate number follows the required format and is not already registered in the system.
7. The system validates that the battery capacity is within a reasonable range (1–200 kWh).
8. If the entered data is valid, the system stores the vehicle record and associates it with the driver's account.
9. The system displays a confirmation message and shows the newly registered vehicle in the driver's vehicle list.
10. The system records the vehicle registration action in the audit log.
11. During a later reservation attempt, the system checks the registered vehicle's connector type against the selected charger's supported connector types and displays the compatibility result.

Outputs:

- Confirmation message indicating successful vehicle registration
- The registered vehicle appears in the driver's profile under "My Vehicles"
- Validation or error messages when input data is invalid
- Compatibility status (Compatible / Incompatible) during later reservation attempts

Exceptions:

E1: Connector type is not supported → System displays error: "Unsupported connector type. Please select Type 2, CCS, or CHAdeMO."

E2: Plate number already registered → System displays error: "A vehicle with this plate number is already registered."

E3: Invalid battery capacity (≤ 0 or > 200) → System displays error: "Battery capacity must be between 1 and 200 kWh."

E4: Missing required fields → System highlights empty fields and displays: "Please fill in all required fields."

E5: Invalid plate number format → The system displays: "Invalid plate number format."

UC-2: Station Search on Map, Reservation Creation, and Route Display

Actors: EV Driver, Map and Navigation Service

Inputs: User's current GPS coordinates (automatic via geolocation), filter criteria (connector type, power output, price range — optional), selected station ID, selected charger ID, selected registered vehicle, desired reservation date and time slot (start time, end time).

Preconditions:

- The EV Driver is authenticated in the system.
- The EV Driver has at least one registered vehicle.
- The Map and Navigation Service is available.

Action Flow:

1. The authenticated EV Driver opens the Map view. The system requests geolocation permission and obtains the user's current coordinates.
2. The system renders an interactive Google Map centered on the user's location and displays charging stations as color-coded markers (green = Available, yellow = Occupied, red = Offline).

3. The driver optionally applies filters such as connector type, power output, and price range. Under normal load, the map updates within 2 seconds to show only matching stations.
4. The system calculates and displays the distance from the user to each visible station.
5. The driver selects a station marker to view station details, including charger list, connector types, power outputs, pricing, and operating hours.
6. The driver selects an available charger and chooses "Reserve."
7. The system prompts the driver to select one of their registered vehicles.
8. The system checks compatibility between the selected vehicle's connector type and the charger's supported connector types. If they are incompatible, the system displays an error message and blocks the reservation.
9. The driver selects a reservation date and time slot.
10. The system validates that the selected slot is within operating hours, does not exceed 2 hours, is not more than 24 hours in advance, and is not already booked.
11. The system calculates the estimated charging cost, displays the estimate and wallet sufficiency status to the driver, and verifies that the wallet balance is sufficient before confirming the reservation.
12. If the reservation request is valid, the system creates the reservation, displays a confirmation message, sends a confirmation notification, and records the reservation action in the audit log.
13. The system displays the route from the user's current location to the selected station using the Map and Navigation Service, including distance and estimated travel time

Outputs:

- Reservation confirmation containing station name, charger ID, reservation date and time, and estimated cost
- Navigation route displayed on the map with distance and ETA
- Confirmation, validation, or error messages shown to the user
- Confirmation notification sent to the user

Exceptions:

E1: Geolocation denied → System displays stations without distance info and prompts: "Enable location services for distance calculation and navigation."

E2: Vehicle-charger incompatible → System displays: "Your vehicle's connector (e.g., CHAdeMO) is not compatible with this charger (e.g., CCS only). Please select a compatible charger."

E3: Time slot exceeds 2 hours → System displays: "Maximum session duration is 2 hours. Please adjust your time slot."

E4: Reservation more than 24 hours in advance → System displays: "Reservations can only be made within 24 hours of the desired time."

E5: Time slot already booked (double-booking) → System displays: "This time slot is already reserved. Please select a different time."

E6: Insufficient wallet balance → System displays: "Your wallet balance (X TL) is insufficient for the estimated cost (Y TL). Please top up your wallet."

E7: Station is offline/under maintenance → System displays: "This station is currently offline. Please select another station."

E8: No registered vehicle → System redirects to vehicle registration: "Please register a vehicle before making a reservation."

E9: User does not start charging within 15 minutes of the reservation start time → System marks the reservation as No-Show, releases the reserved slot, and notifies the user: "Reservation expired, you did not start charging within 15 minutes."

UC-3: Charging Session Processing and Cost Calculation

Actors: EV Driver, Payment / Wallet Service

Inputs: Reservation ID, charging start trigger (user initiates or auto-start at reserved time), real-time energy meter readings (kWh), battery state of charge (SoC %) at start and end, target SoC percentage (optional).

Preconditions:

- The EV Driver is authenticated in the system.
- A valid reservation exists for the selected charger and time slot.
- The charger is operational and available for session start.
- The Payment / Wallet Service is available.

Action Flow:

1. At the reserved time, the authenticated EV Driver arrives at the station and initiates the charging session through the app, or the session starts automatically if auto-start is enabled.
2. The system verifies that the reservation is valid and that the charger is available.
3. The system records the session start time and the initial battery SoC.
4. The system starts the charging session and sends a "Charging Started" notification to the user.
5. During charging, the system continuously tracks and displays real-time session progress, including delivered energy (kWh), current SoC, estimated remaining time, and running cost.
6. The EV Driver monitors the charging session through the application.
7. The charging session continues until one of the following conditions occurs: the battery reaches the target SoC, the maximum session duration is reached, or the user manually stops the session.
8. The system records the session end time and the final battery SoC.
9. The system calculates the total cost based on the total energy consumed and the unit price per kWh.
10. The system deducts the total cost from the user's wallet through the Payment / Wallet Service.
11. The system displays the charged amount, updated wallet balance, and session summary to the user.
12. The system generates a digital receipt containing the station name, charger ID, date and time, start SoC, end SoC, total kWh delivered, unit price, and total cost.
13. The system stores the completed session in the user's charging history.
14. The system records session start, stop, and payment-related actions in the audit log.

15. The system sends a “Charging Completed” notification with the session summary.

16. The system updates the charger status to “Available.”

Outputs:

- Real-time charging-session dashboard showing kWh delivered, SoC, estimated remaining time, and running cost
- Session completion summary
- Digital receipt
- Updated wallet balance
- Charging history entry
- Charging started and charging completed notifications

Exceptions:

E1: Charger malfunction during session → System stops session, calculates cost for kWh already delivered, sends notification: "Charging interrupted due to charger malfunction. You have been charged for X kWh consumed.".

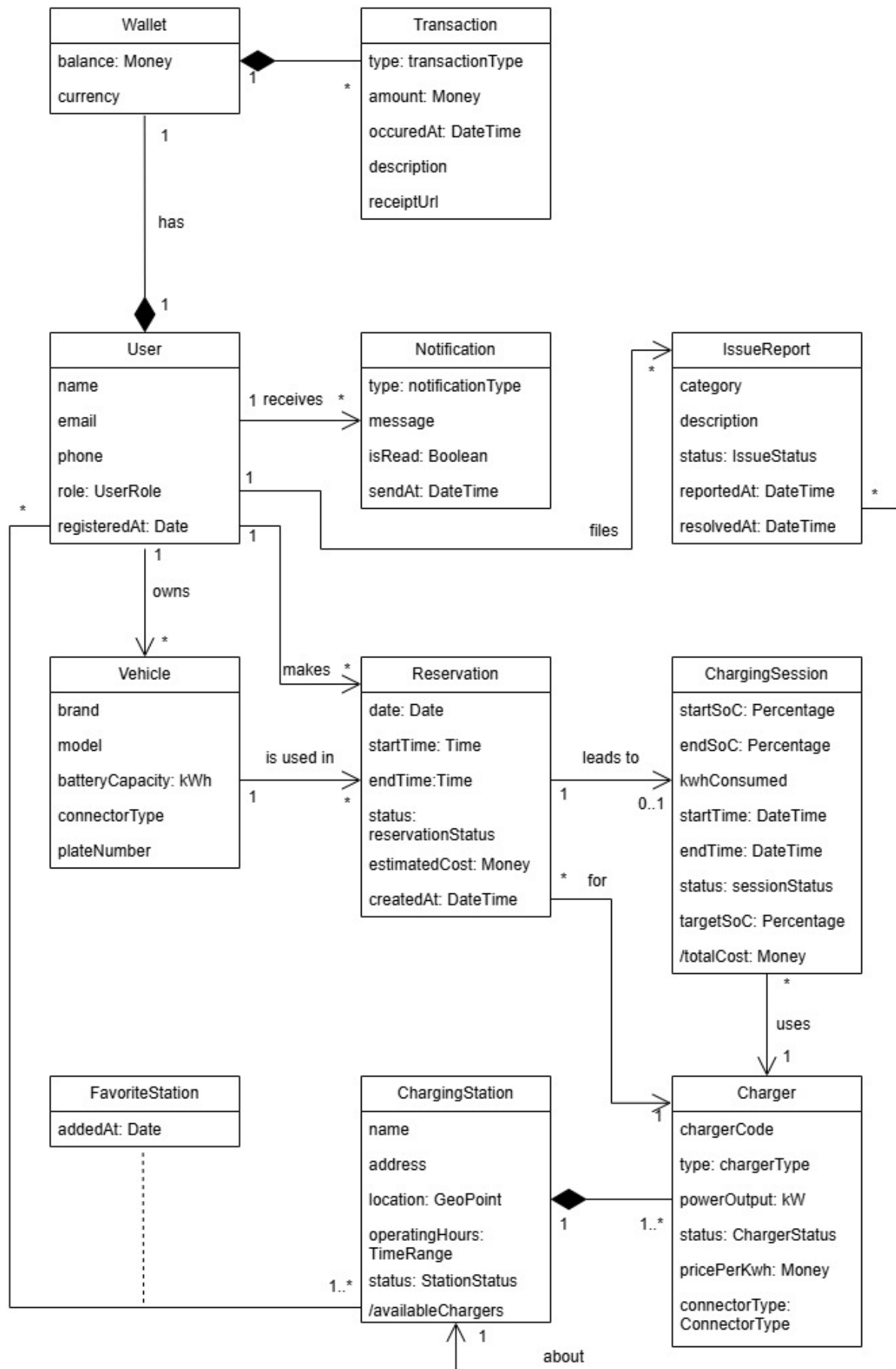
E2: Insufficient wallet balance mid-session → System stops charging safely, deducts the available balance, generates a partial-session receipt, and notifies the user: "Charging stopped — wallet balance depleted. Please top up.".

E3: Session exceeds maximum duration → System auto-stops at 2-hour mark, calculates final cost, notifies user: "Maximum session duration reached. Charging has been stopped.".

E4: Network connectivity loss during session → System caches session data locally and synchronizes cached data within 30 seconds after connectivity is restored. Charging continues uninterrupted.

10. Domain Model (UML Diagram)

The following domain model represents the key entities, their attributes, and relationships within the system.



Entity Descriptions

User: Any system user (EV Driver, Station Operator, or Administrator), distinguished by role.

Vehicle: An electric vehicle registered by an EV Driver. A user can own multiple vehicles.

Wallet: A digital wallet associated with a user, holding the current balance. Each user has exactly one wallet.

Transaction: A financial record (top-up, charge fee, or refund) linked to a wallet. Contains type, amount, description, and receipt information.

ChargingStation: A physical location hosting one or more chargers, with address, geographic coordinates, and operating hours. The number of available chargers is a derived attribute (/availableChargers).

Charger: An individual charging unit within a station, with connector type, charger type (AC/DC), power output, status, and price per kWh.

Reservation: A booking made for a specific charger at a specific time slot, linking User, Vehicle, and Charger.

ChargingSession: An active or completed charging event, recording energy consumed, SoC changes, timing, and status. Total cost is a derived attribute (/totalCost).

IssueReport: A user-submitted report about a problem at a station, including category, description, and resolution status.

Notification: A system-generated alert sent to a user, with type, message, and read status.

FavoriteStation: An association class representing the many-to-many relationship between User and ChargingStation, storing the date the station was added as a favorite (addedAt).

Key Relationships

User → Wallet (1 : 1): Each user has exactly one wallet. Composition — the wallet cannot exist without its user.

Wallet → Transaction (1 : *): A wallet can have many transactions. Composition relationship.

User → Vehicle (1 : *): A user can register multiple vehicles.

User → Notification (1 : *): A user can receive multiple notifications.

User → Reservation (1 : *): A user can make multiple reservations.

User → IssueReport (1 : *): A user can file multiple issue reports.

User ↔ ChargingStation (* : *): Many-to-many relationship via the FavoriteStation association class.

ChargingStation → Charger (1 : 1..*): A station contains one or more chargers. Composition relationship.

Reservation → Charger (* : 1): A reservation is made for a specific charger.

Reservation → ChargingSession (1 : 0..1): A reservation may or may not result in a charging session.

Vehicle → Reservation (1 : *): A vehicle can be associated with multiple reservations, linking it indirectly to charging sessions.

ChargingSession → Charger (* : 1): Each charging session uses exactly one charger. A charger can conduct multiple sessions over time.

IssueReport → ChargingStation (* : 1): Each issue report is about exactly one station.

Enumerations

UserRole: EV_DRIVER, STATION_OPERATOR, ADMINISTRATOR

ChargerType: AC, DC

ConnectorType: TYPE_2, CCS, CHADEMO

ChargerStatus: AVAILABLE, IN_USE, RESERVED, OUT_OF_SERVICE

ReservationStatus: PENDING, CONFIRMED, CANCELLED, COMPLETED

SessionStatus: ACTIVE, COMPLETED, INTERRUPTED, FAILED

TransactionType: TOP_UP, CHARGE, REFUND

IssueStatus: OPEN, IN_PROGRESS, RESOLVED

StationStatus: AVAILABLE, MAINTENANCE, CLOSED

NotificationType: RESERVATION_CONFIRMED, CHARGING_STARTED, CHARGING_COMPLETED, LOW_BALANCE, RESERVATION_CANCELLED

11. Conclusion

This Requirements Analysis Report provides a focused and comprehensive foundation for the EV Charging Station Network Management System. Through viewpoint-oriented analysis, we have identified 63 requirements across eight viewpoints.

Key Findings

Requirement Distribution: The EV Driver viewpoint carries the largest share with 18 requirements, followed by Payment / Wallet Service (9), Administrator (8), Map and Navigation Service (7), Station Operator (6), UI Standards (5), Government Regulations (5), and Security Department (5), for a total of 63 requirements.

Core Functionality: Of the 63 requirements, 48 are functional, 11 are non-functional, and 4 are domain requirements. Together, they cover the complete system workflow, from vehicle registration and map-based station discovery to reservation handling, charging-session tracking, payment processing, security control, regulatory compliance, and administrative reporting.

Stability: 53 requirements (84.1%) are enduring, indicating a stable core design, while 10 requirements (15.9%) are volatile. The volatile requirements mainly relate to configurable policies, evolving UI/design choices, and adjustable security or business rules.

Core Use Cases: Three substantial use cases define the primary system workflows:

- **UC-1 (Vehicle Registration):** Establishes the vehicle registration and compatibility foundation, including authenticated access, input validation, user feedback, and audit logging (EVDR-01, EVDR-02, EVDR-03, UIR-02, UIR-03, SDR-01, SDR-03).
- **UC-2 (Station Search & Reservation):** Integrates map interaction, filtering, compatibility validation, estimated-cost visibility, balance sufficiency verification, no-show handling, cancellation, and route

display (EDR-03 to EDR-08, EDR-13 to EDR-16, MNS-01 to MNS-07, PWS-07, PWS-09, UIR-02, UIR-03, GRR-02, SDR-01, SDR-03).

- **UC-3 (Charging Session Processing):** Covers charging-session start and monitoring, cost calculation, wallet deduction, receipt generation, transaction visibility, charging history, exception handling, and audit logging (EDR-09 to EDR-12, PWS-01 to PWS-08, GRR-01, GRR-02, GRR-04, SDR-01, SDR-03, UIR-02).

Next Steps

- **System Design:** The domain model and requirements will drive architectural decisions and database schema design.
- **Prototype Implementation:** The three use cases will form the core of the prototype demonstration.
- **Testing:** Test cases will be derived from the use case exceptions documented herein.

Statement of Contribution

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